FACULDADE DE
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Advanced Machine Learning
2025/2026 - 2nd Semester

The Butterfly Effect

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1 Introduction

We present here the first practical project, part of the students' evaluation process of the Advanced Machine Learning course of the Master in Artificial Intelligence and Data Science of the University of Coimbra. This work is to be done autonomously by a group of **two students**. The **deadline** for delivering the work is **12th of April of 2026** via Inforestudante. The quality of your work will be judged as a function of the value of the technical work, the written description, and the public discussion. All sources used to perform the work (including the code) must be clearly identified. If you use AI tools in the production of this work (e.g. ChatGPT), you must clearly identify all the parts in which the tool was involved. Please note that, during the defence, you will be required to demonstrate a deep understanding of the content generated by the tool, and this knowledge will be subject to evaluation. The document may be written in Portuguese or in English, using a word processor of your choice[1]. The **written report** is limited to **10 pages long**. The document should be well-structured, including a general **introduction**, **a description of the problem**, the **approach**, the **experimental setup**, an **analysis of the results**, and a **conclusion**. The report should follow the **Springer LNCS format**. The Latex and Word templates are available in the Support Material of the course. The final mark will be given to each member of the group individually. To do the work the student may consult any source he/she wants. Nevertheless, plagiarism will not be allowed and, if detected, it will imply failing the course. While doing the work and when submitting it, you should pay particular attention to the following aspects (whose relative importance depends on the type of work done):

- description of the approach to the problem
- description of the general architecture of the methods used;
- description of the experiment, including a table with the parameters used which should allow full replication;
- description of the evaluation metrics used for the validation: quality of the final result, efficacy, efficiency, diversity, or any other most appropriate;

Do not forget, besides what was just said, that it is fundamental: (1) to do a correct experimental analysis; (2) to do an informed discussion about the results obtained; (3) to put in evidence the advantages of the chosen alternative.

[1] Latex is preferred

2 Problem Statement

Image classification stands as a fundamental problem of computer vision, a field where machine learning has achieved impressive results. Historically, this task relied heavily on human-centric **feature engineering**, where experts meticulously designed algorithms to extract specific visual descriptors for models to process. Once these features were isolated, classical architectures such as *AdaBoost*, *Random Forests*, *Support Vector Machines (SVMs)*, and *Multi-layer Perceptrons (MLPs)* were typically deployed to perform the final classification.

The landscape shifted dramatically with the rise of deep learning, specifically **Convolutional Neural Networks (CNNs)**. While the conceptual foundations of CNNs date back to the late 1980s, their widespread adoption after 2010 was catalyzed by breakthroughs in **GPU-accelerated computing**. Unlike traditional pipelines, CNNs integrate feature extraction and classification into a single, end-to-end learning process. By automatically identifying a hierarchy of patterns – ranging from simple edges to complex organic structures – CNNs have surpassed traditional methods in both accuracy and scalability.

3 Case Study: Biodiversity at the Science Museum

Butterflies represent the adult flying stage of insects within the order *Lepidoptera*, a name derived from the Greek for "scaly wings." This classification is visually literal; their wings are composed of thousands of microscopic, overlapping scales that create the intricate patterns and vibrant colors unique to each species. While these visual markers are essential for survival and mating in the wild, they also provide the high-dimensional features necessary for precise digital identification.

As a lead consultant for the **Science Museum of the University of Coimbra**, you have been tasked with modernizing biodiversity monitoring through an automated computer vision system. Your objective is to design and implement a machine learning model capable of autonomously identifying butterfly species from field-captured images. This system will serve as the technical backbone for the museum's conservation initiative, transforming raw visual data into actionable ecological insights to help protect these delicate pollinators.

4 Objective

The main objective is to analyze the data set and create a machine learning approach with neural networks as the models that can perform supervised image classification of the data set addressed in Section 4.1. To do that you should attend to the following objectives:

- Prepare the machine learning pipeline for the image classification dataset.
- Explore solutions and compare the results using at least, the following ANNs models:
  - Multi-Layer Perceptrons (MLP)
  - Convolutional Neural Networks (CNNs)
  - Residual Neural Networks (ResNets)
- And for each model, experimentation with different layer architectures, loss functions, optimisers and hyperparameters. Explore at least:
  - Loss function: MultiMarginLoss, Cross Entropy
  - Optimizers: RMSprop, ADAM

Exploring other solutions than the listed ones that are suitable for the problem at hand and considered as extra work, can be as compensation points to cover problems in the listed ones above.

4.1 Dataset and Evaluation

The dataset for this project is a comprehensive collection of macro-photographs specifically curated for high-granularity species identification. It features **75 distinct classes** of butterflies, comprising over **5199 labeled images** (inclusive of the validation set). The task is structured as a **multi-class classification problem**, where each image is assigned to a single, mutually exclusive butterfly category.

The data is organized into a hierarchical directory structure. Upon extraction, the **Train** directory serves as the primary repository for model development. This directory contains all the images. The labels are in the train.csv file, including specialized species such as:

- *Southern Dogface*, *Adonis*, and *Brown Siproeta*;
- *Monarch*, *Green Celled Cattleheart*, and *Cairns Birdwing*;

- *Eastern Dapple White*, *Red Postman*, and *Mangrove Skipper*.

Figure 1 illustrates the visual diversity and intra-class variance present in the training set.

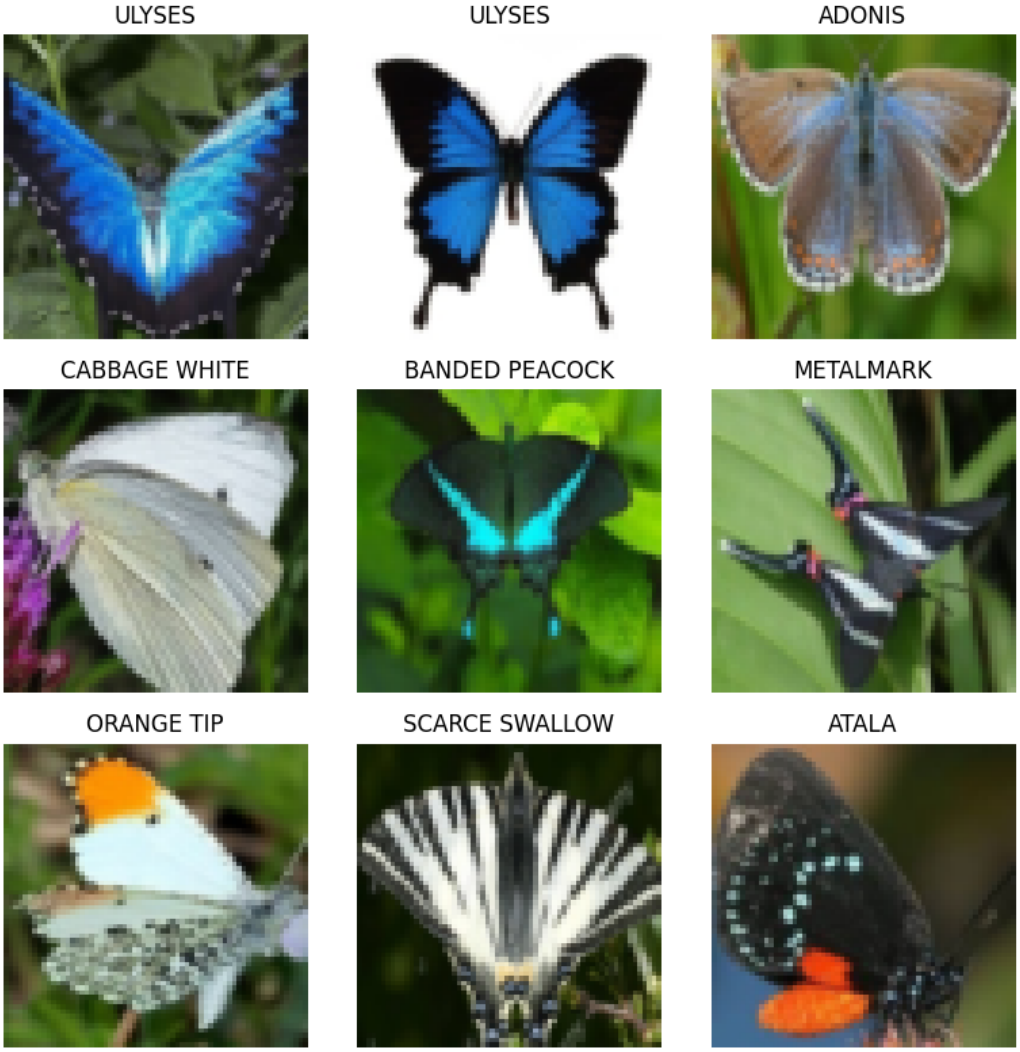

Figura 1: Representative samples from the 75-class Lepidoptera dataset.

As a consultant for the Science Museum, your objective is to leverage this training data to develop a model that generalizes well across all 75 species. You can use the provided script to retrieve and handle the dataset. The main goal is to use the training data to design, implement and validate your approaches, while the test will be used to evaluate the generalization ability of your models. Given the splits you should see how to fit the models that

you are training/creating and select the appropriated metrics to evaluate performance under this context. Thus, the validation part of this work is crucial, and you should select the most appropriate set of metrics and justify them.

5 Competition

To evaluate the generalisation ability of the developed models, we are going to use a Kaggle competition. Note that it will not impact the final mark but rather will act as a way for you to access the progress you are making and evaluate the generalisation performance of your models. The competition is available at the following address:

https://www.kaggle.com/competitions/aca-butterflies/

To participate in the competition, you should prepare a csv file with two columns: the first column contains the Id of the sample that you are classifying, and the second column should contain the corresponding classification label. An example of a submission file is provided along with the project statement.

6 Conclusion

A few short comments. First, the control of the progression of your work will be done during the classes (T and PL). Moreover, you can discuss eventual problems by presenting yourself during office hours. Second, the projects reflect for the most part your current knowledge. The rest will be the object of lecturing soon. Third, we try to balance the difficulty of all the work, but we are aware that this is not an easy task and it is somehow a subjective matter. Fourth, we try to ask for a workload compatible with the value of the work for the final mark.

Methodological issues, like the statistical background, were elucidated during the previous lectures. You may use the statistical tool you feel at ease with, including the Python code that was provided. Finally, even if this is a work that asks you to do simulations and analyse the results, i.e., it has a practical flavour, there is however a theory behind the work, and you are advised to consult the necessary literature.

Good luck!